



UGC Final Year Draft Research Project Proposal

Faculty of Computing

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Study Area and Study Objectives

For many national economies, the food business is crucial because it is one of the largest industries in the world. However, in the next 50 years, there will be a significant increase in food industry / production due to the tremendous rise in global population and the demands placed on the food supply system. In these conditions, significant amounts of food industry wastes acquire national and worldwide social, political and scientific attention.

According to the report of the Food and Agriculture Organization (FAO, 2011a), roughly one-third of food produced for human consumption is lost or wasted globally. Considering the increasing number of hungry people as well as the restricted natural resources, it is easily understood that waste management systems will become one of the most important challenges of the twenty-first century. *(Food Waste Management Using Machine Learning 1 St Vinayak Bharadi, 2 Nd Pavan Jadhav, 3 Rd Omkar Nanche, 4 Th Onkar Munj, 2022)*

Research shows that in Finland, 5% of purchased food is wasted in households, and an average person wasted about 20-30 kg of food in a year. The average total amount of food wasted in households yearly is about 120-160 million kg. (Kale et al., n.d.)

In countries like Finland with a high cost of living, consumers are inclined to buy where the food is near to its expiry date due to the discount shop sellers.

When considering the study objectives

- To identify the main causes of food wastage in various stages of the food supply chain, from production to consumption.
- To determine the primary factors that contribute to food waste at each level of the food supply chain, from production to consumption.
- To examine the current food donation systems in the study area, but considering their strengths, weaknesses and identifying the opportunities that we can increase.
- To formulate suggestions for enhancing the efficiency of food donation programs and lowering food waste in the research region.

Implementing a donation program and management system for food waste has several advantages. They comprise:

- Benefits for the environment: Cutting down on food waste helps to preserve natural resources, cut back on greenhouse gas emissions, and reduce the environmental effects of food production, distribution, and disposal.
- Social advantages: Donating extra food can help local communities, especially vulnerable ones like low-income families, kids, and the elderly, combat hunger and food insecurity.

- Economic benefits: By decreasing food waste and donating surplus food to charities, businesses can save money on trash disposal costs and may be eligible for tax breaks or other rewards for charitable donations.

Motivation for selecting the Research Area

Many people waste food nowadays across the entire world and because of the inflation many NGO's, animal farms, elderly households don't get food. The United Nations estimates that over 1.3 billion tons, or about one-third, of the food produced worldwide is wasted annually. Additionally, millions of people worldwide experience food insecurity and hunger. This motivates me to do a research on food wastage occurring from hotels, guest houses, etc.. as in many food wastages occurs in those places and from those places if we could be able to donate those food for the places through a management system it would be effective to the whole world. The implementation of a food waste and donation system can effectively address issues such as hunger, sustainability, and environmental impact. By donating excess food to those in need, we not only provide immediate relief to the hungry but also contribute to the reduction of food waste that would otherwise end up in landfills. Furthermore, the reduction of food waste can lead to resource conservation, including water, energy, and land.

Working Topic

So based of the current research analysis I have done, globally food wastage is a huge problem and because of inflation of the world many people and animals are suffering from hunger so I have thought of researching about food wastage and how can we donate those wastage foods to the NGO's and animal farms from hotels, restaurants etc....

Research Gap

In the literature review I have got to identified that this system is an existing system with lack of innovative features and also according to the research knowledge I have gathered there only one mobile application performing properly just only for the delivery purposes.

To develop a Web Based application that reduces the amount of food wastage produced in restaurants, functions and mess. The current system only provides information on amount of food wasted and does not provide an interface to donate and provide data analysis. Using data analysis, to visualize the impact. Donating the excess food that consists of the following details, first, providing the location of where excess food is available & details of the food quantity available. (Kale et al., n.d.)

A web application that is to create and assist users in managing their food inventory. The application will be store and display basic information about the inventory contents and also alerts the user of the food products which are due to expire the next day. (Monica M & DharaniNV, n.d.)

However, the application is limited Android Smart phones with Gingerbread OS and higher versions. Also, the application will be beneficial if donors and seekers are located near each other. The application has 3 actors: - Donor, Receiver and Admin. The donor can view Receiver's location and receiver can view donor's location since it is necessary for the delivery.(*Food Waste Management Using Machine Learning 1 St Vinayak Bharadi,2 Nd Pavan Jadhav,3 Rd Omkar Nanche,4 Th Onkar Munj, 2022*)

According to the literature reviews and research I have done I have mainly identified that in many management systems there are no delivery services as in the NGO should go and take the food they need and many restaurants won't donate foods for farms as well. Here delivery people can use the same website / application to get registered and he can accept orders when the order is placed. The delivery people are assigned by the NGO's and Animal Farms. When the order is accepted from the people who get food with the details a QR code will be generated and they can share the QR code along with the delivery person who goes to get the food from the hotels. The main gap between the existing website and the ideas I have get is the mobile application as in that would be easy for the donors , drivers and the customers.

Research Problem / Research Questions

According to the literature review I have done I have identified that there are globally affected problems in this food sector. Food waste is any product that is disposed after they has been prepared, processed, sold but there are several factors including excess production, food degradation and customer preferences for a perfect produce. Around the whole world many people suffer from hunger and malnutrition not only people but many animals also suffer from many disease due to lack of food. But the main problem Is many hotels, restaurants they won't donate food because of the transportation cost, storage and distribution cost and food safety regulations. So the problem is to make the hotel , restoration etc.. owner to donate food without wasting them in an effective manner and in a very easy way with ensuring that the surplus won't get affected to the owners and by ensuring the food security.

When we considering the research questions →

- How can technology get involved in the donation process of surplus food and to reduce the food wastage?
- What can we do to get the society know about the food wastage and the benefits that can be gain through donating food for NGO's and animal farms?
- What are the main factors that food gets waste in houses and how can these waste food can be reduce without throwing them ?
- How can retailers better manage their inventory to reduce food waste and increase opportunities for donation?

Research Strategy / Research Model

Along with the literature review I have done and the strategy problem I have found the best way for the research strategy is doing a survey to identify about the problem I have found. Surveys are a most popular research strategy in the whole world as in they allow the researchers to collect the data what we need to from a larger no of participants within a short period. And its very flexible and collect a wide range of data including many areas. (Demographic information, attitudes, opinions, and preferences). Overall surveys are a most powerful research strategy that can provide a wide range of valuable insights into the attitudes, opinions, and behaviors of the target population.

Surveys can be conducted with different types of models such as, questionnaires, online surveys, phone surveys, and interviews. Depending on the research I'm considering to choose online surveys and in-person interviews. It would be really affecting for my research as the research problem affects to the whole county and world. When analyzing online surveys, I have thought of using a google form for the data collection of the research and in-person interview I have targeted the people between (25 – 40) as these age range people are aware of this problem.

Research Timeline

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